

Senior front-end developer specialising in Angular with a wealth of experience from working at a digital agency.

I lead front-end development across two teams, one delivering an enterprise web application for a FTSE 250 client, the other serving a high-growth startup that I have had the privilege of working on from MVP through to launch.

This startup is now turning a profit with revenues over £1 million and still growing. This has given me experience of the different stages of a product's lifecycle and the different emphases and responsibilities that those stages entail.

Skills

- Collaborating with designers, back-end developers, data engineers, product owners and stakeholders to ensure continuous delivery of high-quality software.
- Leveraging **Angular**, **NgRx**, and **RxJS** to build enterprise SPAs that remain performant, secure, and maintainable across time while integrating with **C# .NET** and or **GraphQL** APIs.
- Building Angular applications with **server-side rendering** to maximise **SEO** performance.
- Using W3C compliant **HTML**, **CSS**, & **JS** to deliver cross-browser support, responsive design, and accessibility for our users.
- Unit testing, integration testing, and **E2E testing** with **Jest/Karma/Playwright**.
- Application instrumentation and monitoring via **Azure**, **Application Insights**, and **OpenTelemetry**.
- Setting up **Google Analytics** with **GTM** to monitor user behaviour and generate insights for the business.
- Maintaining a highly reusable and customisable **component library** for use across the whole business.
- Integrating with **CMS** (headless & traditional), as well as payments, messaging, and authentication libraries.
- Inventing novel solutions to complex problems that sit outside the typical frameworks when the need arises.

Professional Experience

Rockpool Digital (3 years)

Rockpool is a digital agency delivering bespoke web applications for some of the UK's biggest brands.

Senior Front-End Developer Sep 2023 - Present

Lead two teams of front-end developers, overseeing architecture and best practices across our applications. Chair team meetings and mentor and support developers through regular 1:1s. Assist stakeholders and product owners in shaping the product roadmaps, advising on technical scope and risk mitigation.

Key contributions:

Created a new SPA with Angular 18 and an integration with Umbraco's headless CMS that enabled the growth team to double the click-through rate to our quote engine; the SPA also utilised server-side rendering and Azure Front-Door caching to improve SEO performance.

Upgraded our approach to application logging across our apps, utilising Azure's Application Insights and wiring it up to configcat-js, our feature flagging service. This enabled us to fine-tune the logging and sampling rate on the fly, targeting specific user groups if desired, while also controlling costs.

Eliminated errors that were impacting users through continuous monitoring of the application logs with a set of custom KQL queries. Also managed to weed out a lot of the less impactful exceptions that usually stemmed from poor architectural approaches or shaky integrations with 3rd party dependencies.

Reimplemented our consent management solution, leveraging Google Tag Manager (GTM) to integrate with a consent management platform (CMP) that ensures full compliance with all relevant privacy laws while simultaneously tracking opt-outs for better insight into overall traffic patterns.

Architected a Node.js Express server built on top of an Angular app that I configured to render MJML markup rather than HTML. The express server renders the generated MJML into fully rendered email templates to be consumed by our BE APIs. This eliminates the need for the BE services to create and maintain fiddly HTML templates and allowed the growth team to self-serve for campaign emails.

Created a suite of integration tests for a mission-critical feature utilising Jest and pixelmatch for pixel-by-pixel comparison of PDF output to ensure consistency and reduce regression bugs across changes that were previously resulting in frequent unintended consequences.

Defined and documented our ideal FE architecture that we now aim for across all our Angular applications. For a long time we had diverging emergent architectures across our applications that increased resistance to team members moving between projects. With the new architecture understood and accepted by the whole team, expectations are much clearer, and confidence in what we are delivering has improved.

Angular Developer Oct 2022 - Aug 2023

Led FE development for our largest client, delivering big changes and upgrades to an enterprise-scale Angular application that had suffered from a lack of FE expertise in the years prior.

Key contributions:

Made a huge impact by delivering a 5x performance gain to multiple application KPIs and unlocking previously unreachable functionality. The application was designed to streamline the creation of property valuation reports via an in-app editor backed by an HTML-to-PDF 3rd party dependency. However, the time it took to generate these reports was intolerably slow for use within an edit-preview-edit loop of a non-WYSIWYG editor.

By writing a custom JS framework for PDF generation, I was able to improve performance across section previews and full report generations by a factor of 5 while also pushing compute out to end-user devices, reducing strain on the BE server. The performance of this framework scaled linearly with the size of the input, ensuring scalability, and also unlocked fine-toothed control over the look and layout of the complex reports being produced.

Pushed for change in the way we used NgRx for state management. It had been bolted on to the project, and the usage was full of anti-patterns, resulting in buggy and unreadable code. After leading some workshops and showing the way forward with some big refactoring efforts, we saw a dramatic fall in developer stress scores for that particular team and a big reduction in missed deadlines and regression bugs, earning myself the *Change Catalyst* quarterly award.

Led the delivery of a complex front-end integration with government APIs for statutory data such as title boundaries, planning permissions, EPC data, and listed building status. This data was displayed out to users using MapBox GL for visualisation and selection of title boundaries on a map view.

Junior Front-End Developer Jan 2022 - Sep 2022

Delivered features, 3rd-party integrations, and ticket resolutions, exceeding expectations and earning myself the *Team Player* quarterly award during my first quarter at the company.

Education

M.Sci Computer Science (1st class, 4.0 GPA) 2016 - 2021
University of Birmingham